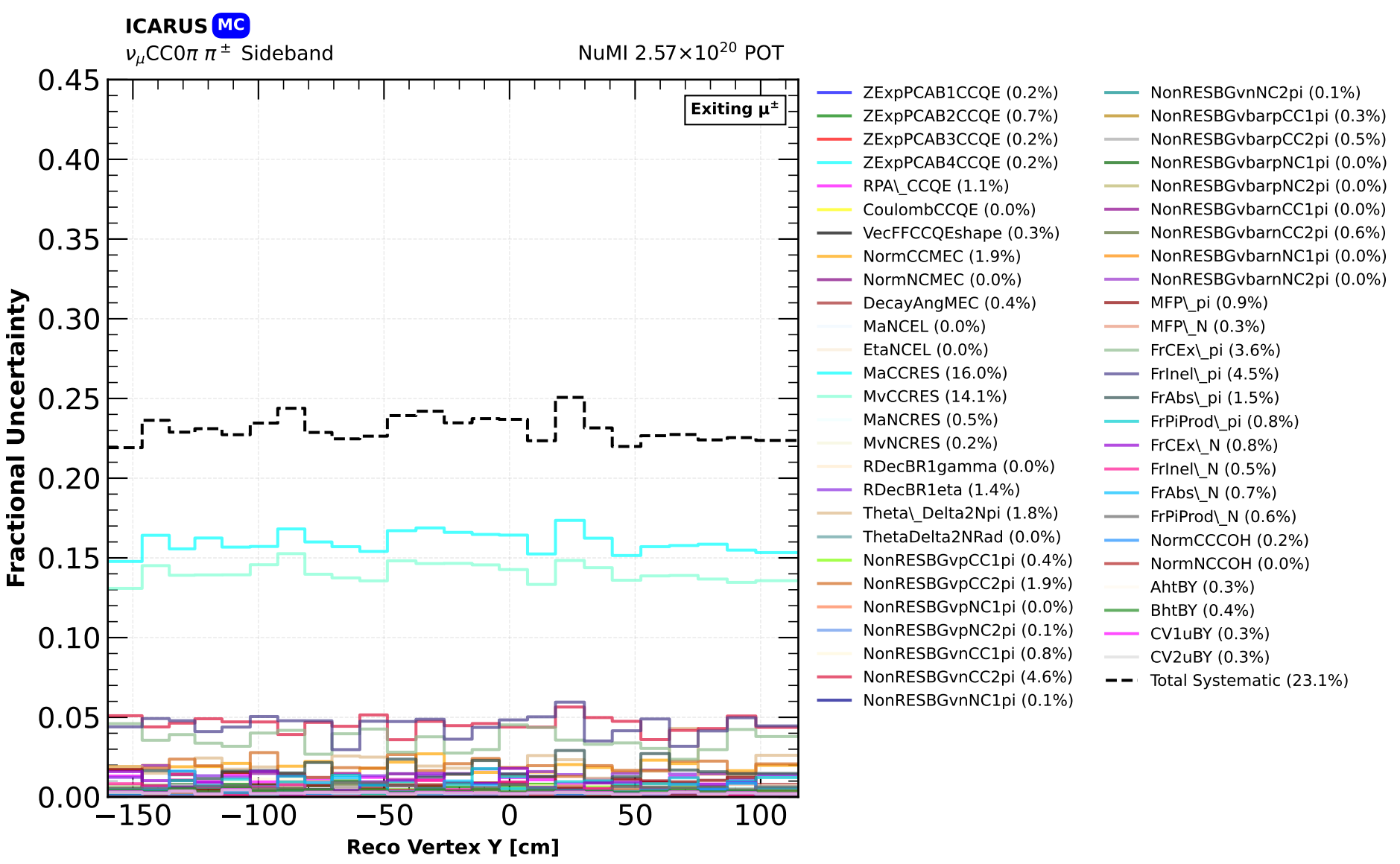
ICARUS MC
ν_μCC0π $\pi^\pm$ Sideband
NuMI 2.57×10^20 POT
Exiting μ±
Fractional Uncertainty
Reco Vertex Y [cm]
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−150
−100
−50
0
50
100
ZExpPCAB1CCQE (0.2%)
ZExpPCAB2CCQE (0.7%)
ZExpPCAB3CCQE (0.2%)
ZExpPCAB4CCQE (0.2%)
RPA_CCQE (1.1%)
CoulombCCQE (0.0%)
VecFFCCQEshape (0.3%)
NormCCMEC (1.9%)
NormNCMEC (0.0%)
DecayAngMEC (0.4%)
MaNCEL (0.0%)
EtaNCEL (0.0%)
MaCCRES (16.0%)
MvCCRES (14.1%)
MaNCRES (0.5%)
MvNCRES (0.2%)
RDecBR1gamma (0.0%)
RDecBR1eta (1.4%)
Theta_Delta2Npi (1.8%)
ThetaDelta2NRad (0.0%)
NonRESBGvpCC1pi (0.4%)
NonRESBGvpCC2pi (1.9%)
NonRESBGvpNC1pi (0.0%)
NonRESBGvpNC2pi (0.1%)
NonRESBGvnCC1pi (0.8%)
NonRESBGvnCC2pi (4.6%)
NonRESBGvnNC1pi (0.1%)
NonRESBGvnNC2pi (0.1%)
NonRESBGvbarpCC1pi (0.3%)
NonRESBGvbarpCC2pi (0.5%)
NonRESBGvbarpNC1pi (0.0%)
NonRESBGvbarpNC2pi (0.0%)
NonRESBGvbarnCC1pi (0.0%)
NonRESBGvbarnCC2pi (0.6%)
NonRESBGvbarnNC1pi (0.0%)
NonRESBGvbarnNC2pi (0.0%)
MFP_pi (0.9%)
MFP_N (0.3%)
FrCEx_pi (3.6%)
FrInel_pi (4.5%)
FrAbs_pi (1.5%)
FrPiProd_pi (0.8%)
FrCEx_N (0.8%)
FrInel_N (0.5%)
FrAbs_N (0.7%)
FrPiProd_N (0.6%)
NormCCCOH (0.2%)
NormNCCOH (0.0%)
AhtBY (0.3%)
BhtBY (0.4%)
CV1uBY (0.3%)
CV2uBY (0.3%)
Total Systematic (23.1%)